

ASSIGNMENT – IV

C language: Control statement and Arrays

4.1 A positive integer is entered through the keyboard. Along with it the base of the number system in which you want to convert this number is entered. Write a C program to do this conversion.

4.2 Write a C program to print all numbers between 1 to 100 which divided by a specified number and the remainder will be 3.

4.3 Write a C program Write a program to print each of the following output patterns.

```
*           * * * * *           *
* *         * * * * *         * * *
* * *       * * *           * * * * *
* * * *     * *           * * * * * *
* * * * *   *           * * * * * * *
1           1
0 1         2 2 2
0 1 0       3 3 3 3 3
1 0 1 0     4 4 4 4 4 4 4
1 0 1 0 1   5 5 5 5 5 5 5 5 5
```

4.4 Write a C program to print the following Pascal's Triangle.

```
1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
```

- 4.5 Write a C program to print the elements of an array in reverse order.
- 4.6 Write a C program to find the largest and smallest elements of an array.
- 4.7 Write a C program to swap the elements of two arrays.
- 4.8 Write a C program to find the average of the elements of an integer array. Also find the standard deviation of these elements.
- 4.9 Write a C program to sort a set of n numbers in descending order and then search for a given number (taken from user) in the set of sorted numbers.
- 4.10 Write a C program to insert an element (given by the user) into an array in a particular position (given by the user).
- 4.11 Write a C program to delete an element from a particular position (given by the user) in an array.
- 4.12 Write a C program for shifting an element of an array one location left.
- 4.13 Write a C program to print Fibonacci series using array.